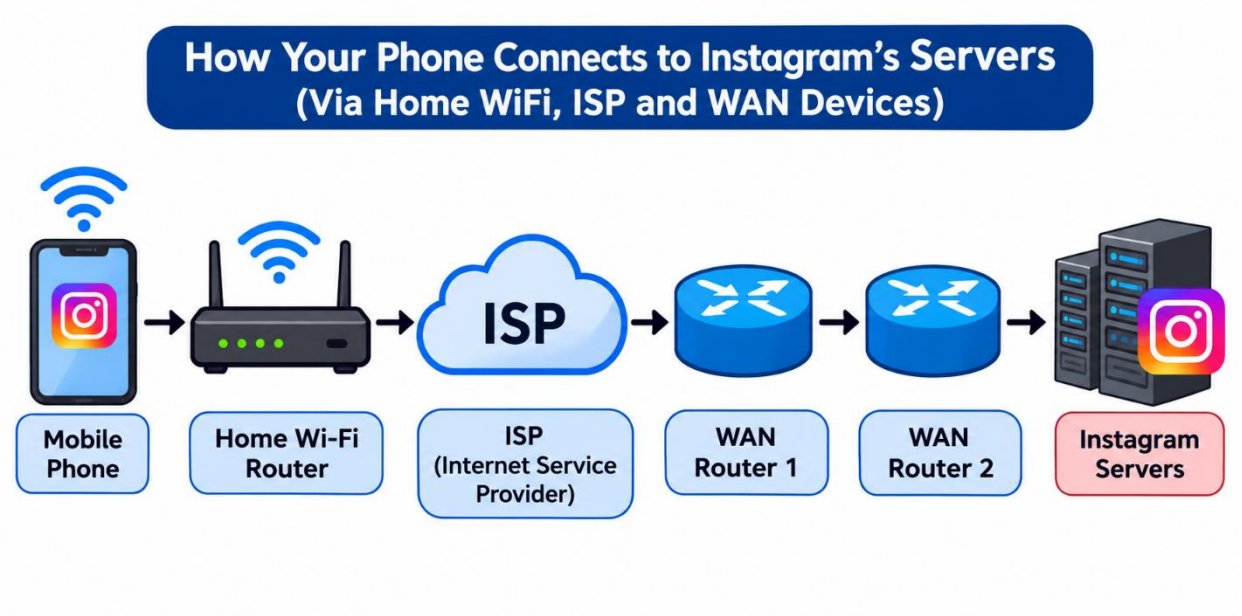
Assignment – 2.5

Wide Area Networks

Q - 1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



Q - 2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

➢ **Comparison of WAN Technologies:**

WAN Technology	Speed	Cost	Real-Life Usage
MPLS	High speed and reliable	High	Zomato offices connecting different branches securely.
Leased Line	Very high and dedicated speed	Very high	IRCTC ticketing system for reliable communication and data transfer.
Broadband Internet	Medium to high speed	Low	Home Wi-Fi used for browsing, YouTube, Instagram and online classes.

<u>Q - 3.</u> Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Hint: Focus on how the VPN creates a private tunnel over the public WAN.

- **<u>How VPN Uses WANs:</u>**
  - A **VPN (Virtual Private Network)** creates a secure tunnel over a public WAN (internet).
  - For example, when we use public WiFi at a coffee shop to watch YouTube or listen to Spotify, a VPN protects our data by encrypting it.
  - It keeps our personal information safe from hackers using the same WiFi.
- **<u>In short:</u>** VPN makes public WiFi safer by creating a private and secure connection.

Q - 4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

- **Best WAN Technology: MPLS**

  Flipkart should use **MPLS (Multiprotocol Label Switching)** to connect its warehouses in Ahmedabad, Mumbai, and Delhi.

- **Two Reasons:**

  1. **High Speed:** MPLS provides fast and reliable communication between all warehouses.

  2. **Security:** It provides a secure network to share important business data, such as stock details and delivery information.